



Research Paper

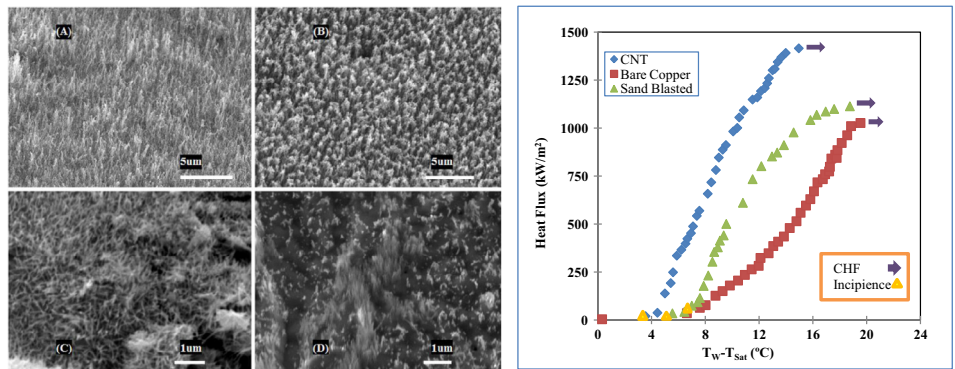
Pool boiling heat transfer enhancement using vertically aligned carbon nanotube coatings on a copper substrate

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HIGHLIGHTS

- Vertically aligned carbon nanotubes were coated on copper substrate.
- Diamond is used as the intermediate layer to reduce the intermediate resistance.
- Pool boiling heat transfer studies were conducted.
- A 38% enhancement in critical heat flux was achieved using vertically aligned carbon nanotube coating.
- Shift of boiling curve toward the left due to the wettability and fin action.

GRAPHICAL ABSTRACT



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ABSTRACT

In the present study, pool boiling heat transfer experiments are performed on a bare copper surface, a sandblasted copper surface, and carbon nanotube coated copper substrates. Experiments were conducted on the substrate surface area of $14 \times 14 \text{ mm}^2$ using de-mineralized water as the working fluid. The substrate is fixed tightly on the heater block, which is embedded with six cartridge heaters of total capacity 1500 W. Heat fluxes were measured using J-type thermocouples, which were aligned vertically along the heater block and also 1 mm from the top of the substrate. The pool boiling heat transfer experiments were conducted on bare, sandblasted and vertically aligned CNT coated copper substrates. The obtained results of sandblasted and CNT coated surfaces were compared with that of the bare copper surface. A remarkable enhancement of 38% in the CHF was observed on the CNT coated surface. The CNT coated surface was found to be effective in increasing the nucleate boiling and the CHF.

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1. Introduction

Heat transfer enhancement is the one of the major concerns in heat exchangers and micro electro mechanical system (MEMS). The recent advancements in technology have contributed to a tremendous size reduction of electronic devices, which has led to a high

heat flux. So, the cooling of these devices has become a challenging task. Among the different heat transfer techniques in electronic devices and microprocessors, pool boiling is an effective method due to its change of phase. Different passive techniques are used to enhance heat transfer such as increase of sub cooling, increase of surface roughness by surface modification and surface coating [1]. Among the different passive techniques, surface coating is an efficient method to improve the heat transfer characteristics by increasing the effective heat transfer area, nucleation density, bubble frequency and wettability [2–5].

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Surface coatings can be macro-structured, micro-structured, and nano-structured; among them, nano-structured coatings have potential application in MEMS devices. Nano-structured coating possesses the following desired properties: high surface area to volume ratio, smaller volume and mass, and high convective heat transfer coefficient; these causes the removal of large amounts of heat flux. Porosity, fin action and surface roughness of the nanostructured coating are the major parameters which enhances the heat transfer [6–13]. Extensive research has been carried out on pool boiling using surface coating. Ahn et al. conducted flow boiling and pool boiling analyses on different metallic substrates coated with TiO_2 , SiO_2 and Ag nano particles. For pool boiling and flow boiling, a significant increase in the critical heat flux (CHF) was recorded for all coated surfaces. The enhancement of CHF on coated surfaces is mainly due to change in wettability and capillarity which increase liquid spreading and actively supplied cooling water to the heating surface. Sujith Kumar et al. [14] conducted the flow boiling heat transfer experiments and found that 21.6% of CHF was enhanced at the flux rate of $283 \text{ kg/m}^2\text{s}$ over carbon nanotubes with diamond coating acting as intermediate layer on a copper substrate for the demineralized water as a working fluid. Similar kind of surface coating was thought to be effective in case of pool boiling and is tried in this work. The investigation may lead to some deeper insights into the actual performance of vertically aligned CNTs for the enhancement of phase change heat transfer. This was the drive behind this investigation in the case of pool boiling heat transfer leading to applications like electronics cooling. Mudawar and Anderson [3] found out a pool boiling augmentation on finned and microstudded surfaces. These surfaces are more effective to enhance the CHF due to increase in surface area with a mild increase in the fin volume. Ahn et al. [6] conducted pool boiling experiments on CNT coated silicon substrates. Experiments were conducted on bare silicon wafer and two silicon wafers, each deposited with a uniform layer of vertically aligned MWCNTs of different heights of $9 \mu\text{m}$ and $25 \mu\text{m}$ respectively. A maximum enhancement of 40% in CHF was obtained for the $25 \mu\text{m}$ height CNT coated surface. Ujereh et al. [4] conducted pool boiling experiments for assessing the impact of CNT coating on both copper and silicon substrates. Experiments were conducted using FC-72 as the working fluid. They obtained 35% of enhancement in CHF for the coated substrate than for the bare substrates.

The recent advances in nanotechnology have led to the development of carbon nanotube (CNT) coatings over metallic substrates, which are the most efficient way to remove the high heat flux from small areas. Pathak et al. [15] coated the CNT fins over the shape memory alloy wire, and conducted a study on heat transfer. They found an appreciable enhancement in the cooling rate of the CNT coated wire compared to the uncoated one. The enhancement in the cooling rate was due to the high thermal conductivity of the CNT and increase in the total heat transfer area.

Jean-Marc Bonard et al. [16] stated that a major problem for CNT coatings with the metallic substrate is the poor adhesion between the CNT and the substrate, which will increase the intermediate resistance. Extensive research is being carried out to overcome this problem. The adhesion between the metal and the CNT can be increased by using an intermediate layer. Kumar et al. [17] used omega bond gluing agent as an intermediate layer to increase the adhesion between the CNT and the stainless steel substrate. Song et al. [18] reported that adding a thin layer of indium improved the adhesion between the CNT arrays and the substrate. Khanikar et al. [19] conducted a flow boiling heat transfer studies on CNT coated copper substrates. They prepared the CNT coating over the copper substrate, with a tri-layer of titanium, aluminium and iron as the intermediate metal catalysts. The titanium layer gives high bonding strength for the CNT, the aluminium layer was used for segregating the annealed iron nano particles, and the iron layer acts as a

catalyst for the CNT growth. Lin et al. [20] used the Al_2O_3 intermediate layer as a thermal interface material to coat the vertically aligned CNTs on the copper substrate.

In the present work, to improve the adhesion between the CNTs and the Cu substrate, a diamond intermediate layer was coated between the CNTs and the copper. Furthermore, to enhance the adhesion between the diamond layer and the copper substrate, a diamond layer deposition was achieved on a sandblasted copper substrate. The hot filament chemical vapour deposition technique is used for coating the diamond and the CNT. Pool boiling heat transfer studies were conducted on the CNT coated copper substrate, bare copper and sand blasted copper substrate.

2. Experiment

2.1. Double-layered structure deposition

The oxygen-free copper substrate was etched with HNO_3 and HCl , and then ultrasonic cleaning was carried out in a bath containing 99% ethanol for 30 minutes. The cleaned copper substrate was sand-blasted for 1 minute, and prepared for the HFCVD deposition. Copper substrate was first ultrasonically seeded in a suspension of ethanol and nano-size diamond powders, and placed in the HFCVD chamber for the deposition. The precursor gases are CH_4 (2 vol. %) and H_2 (98 vol. %) with a total flow rate of 50 sccm and the pressure inside the chamber was maintained at 20 torr at 400°C substrate back temperature. For CNTs growth: a ferric nitrate catalyst was coated on the diamond layer by spin coating, and the spin coating time was set at 30 seconds. A dc-bias voltage of -500 V and a glow discharge current of 100 mA was supplied to assist the CNT alignment growth in a CH_4 (20 vol. %) and H_2 (80 vol. %), with a gas mixture of total gas flow rate of 50 sccm.

2.2. Details of characterization equipment

The surface morphology of the Cu substrate and diamond coatings was observed using SEM (JSM-6010). The surface roughness was measured by MitatoyoSufstest (ISO 1997 standard, cut off length 20 mm, resolution: $0.0001 \mu\text{m}$). The phase analysis was done by the Raman Microscope (Renishaw 2000) and XRD (RigakuGeigerflex XRD). Static contact angle of the surface was measured by contact angle metre (Holmarc, HO-IAD-CAM-01A).

2.3. Experimental apparatus

The constructional details of the experimental setup used in this study are shown in Fig. 1. The experimental setup consists of the following components: 1) a viewing chamber, 2) test plate 3) heating block, 4) auxiliary heater, 5) reflux condenser, 6) insulation, 7) pressure transducer, 8) data logger and j type thermocouples are used where temperatures to be measured and the viewing chamber are made of acrylic enclosure. The copper test plate is heated by using six cartridge heaters with a capacity of 250 W each. The rate of heat is controlled by an autotransformer of the range of 0–2000 W. An auxiliary heater was provided for controlling the fluid bulk temperature, and a water cooled condenser is used for condensing the vapour produced during the pool boiling process. A thick layer of glass wool insulation was provided for reducing the heat loss to the surroundings. Heat loss in the radial direction in the heater block was analyzed using COMSOL multiphysics software and was found to be negligible. This result was well expected as the heater block was well insulated radially using Teflon and glass wool, thus minimizing the heat loss to negligible values. Three J type thermocouples were inserted at 1 mm below the top surface of the copper substrate for measuring the surface temperatures along the length of the substrate and two J type thermocouples were inserted 4 mm

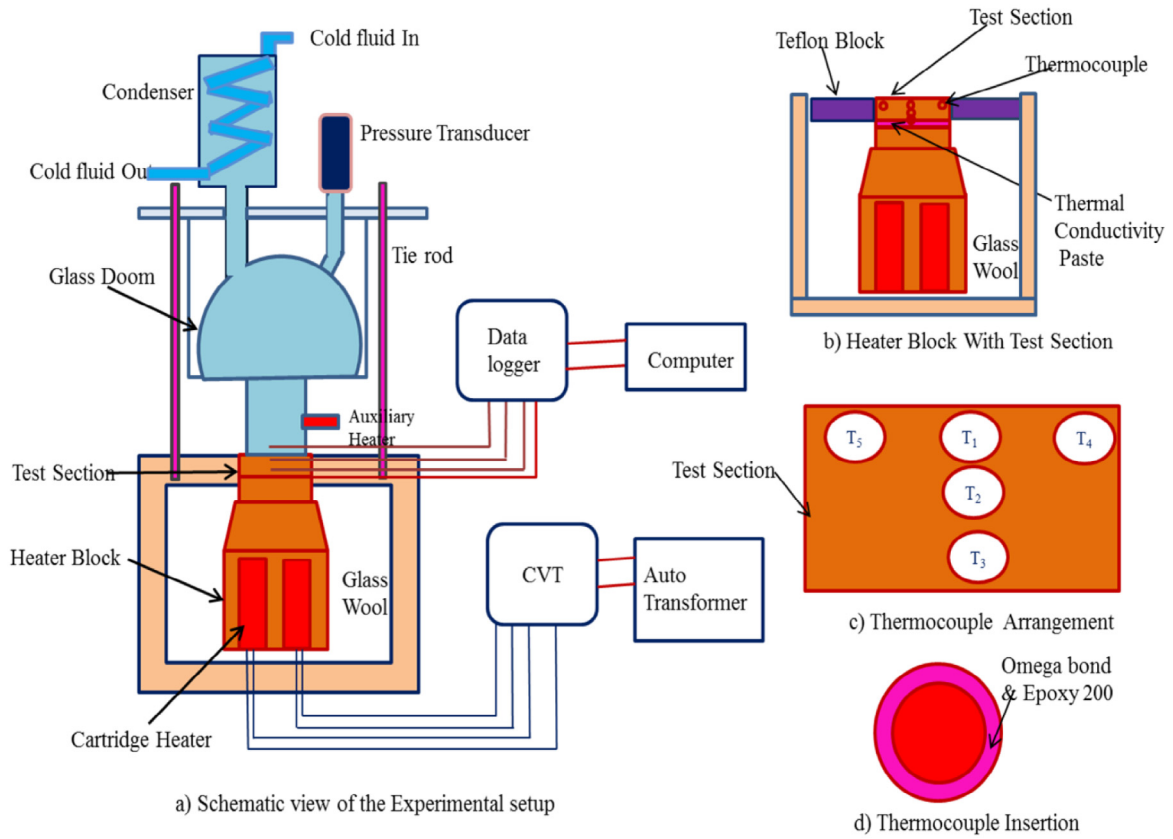


Fig. 1. Schematic of experimental setup and arrangement of thermocouples to measure heat flux.

apart in the copper substrate along the vertical section. The bulk temperature of the fluid is measured by using a 1 mm diameter J type thermocouple. The instantaneous temperatures measured by the thermocouples were digitalized using a data logger (data taker dt85). All thermocouples were calibrated using constant temperature oil bath with a calibrated digital thermometer. The maximum error in the temperature measurement was found to be $\pm 0.5^\circ\text{C}$. All uncovered parts of heating section were insulated with a thick layer of glass wool and Teflon block.

2.4. Data analysis

$$q = -k \frac{T_2 - T_1}{\Delta x} \quad (1)$$

The experimental wall heat flux data are obtained by calculating the temperature gradient in the vertical direction of the substrate.

The boiling heat transfer coefficient can be calculated from the heat flux, the average surface and the average fluid temperatures using the formula given below.

$$h = \frac{q}{T_s - T_f} \quad (2)$$

2.5. Experimental uncertainty analysis

The thermocouples were calibrated using a digital thermometer and a constant temperature oil bath. The thermocouples were calibrated by maintaining the constant temperature bath in the steady state at different temperatures, in the operating range of the thermocouples. The uncertainty in the temperature measurement was $\pm 0.5^\circ\text{C}$. The uncertainty in position was found to be $\pm 2.1\%$, and

that of thermal conductivity was $\pm 1\%$. The experimental uncertainty in the heat flux in the vertical direction in the copper test section, and the boiling heat transfer coefficient were estimated by the Kline and McClintock method [21].

$$\frac{\omega_q}{q} = \left[\left(\frac{\omega_k}{k} \right)^2 + \left(\frac{\omega_{T1}}{T_2 - T_1} \right)^2 + \left(\frac{\omega_{T2}}{T_2 - T_1} \right)^2 + \left(\frac{\omega_{\Delta x}}{\Delta x} \right)^2 \right]^{1/2} \quad (3)$$

$$\frac{\omega_h}{h} = \left[\left(\frac{\omega_q}{q} \right)^2 + \left(\frac{\omega_{T_s}}{T_s - T_f} \right)^2 + \left(\frac{\omega_{T_f}}{T_s - T_f} \right)^2 \right]^{1/2} \quad (4)$$

The experimental uncertainty for the wall heat flux and boiling heat transfer coefficient was estimated to be in the range of $\pm 3.57\%$ and $\pm 2.35\%$ respectively.

2.6. Experimental procedure

Pool boiling experiments were conducted by using de-mineralized water. Prior to the experiment, the water was boiled vigorously for 30 minutes to remove the dissolved gases. The bulk temperature of the water was maintained constant by using an auxiliary heater. The heat was supplied to the testing plate in small increments by six electrical cartridge heaters inserted in a copper block. Once the temperature in the copper block reached the steady state, all the temperature data from the thermocouples were recorded by using a data logger. After all the measurements were taken, the heat flux of the testing plate was increased, and the measurement procedure was repeated. The experiment was repeated for different values of heat flux until the CHF is encountered which can be identified by an abnormal rise in the surface temperature [6].

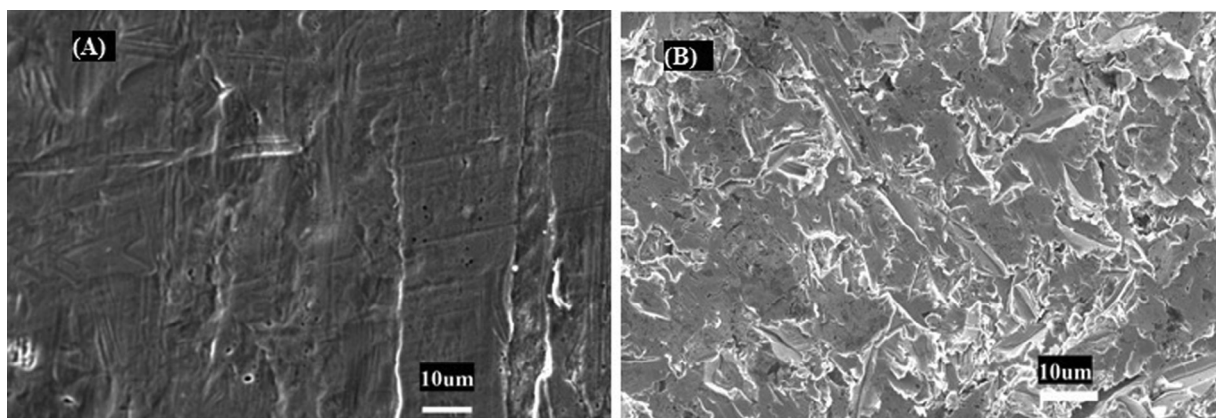


Fig. 2. SEM images of the (A) bare copper surface and (B) sand blasted copper.

3. Result and discussion

3.1. Characterization of the coating

Fig. 2 shows the SEM image of the surface morphology of untreated (A) and sandblasted (B) Cu substrate. It can be observed that the surface roughness has significantly increased after sandblasting. The surface roughness measurement results in Table 1 can prove it. Fig. 3 shows the SEM image of diamond coatings on untreated Cu substrate (A) and sandblasted Cu substrate (B). The diamond coatings were partly peeled off from the untreated Cu substrate and were stayed on sand blasted Cu substrate. Diamond and Cu have a large thermal expansion mismatch. When a diamond coated on Cu cools down from a higher deposition temperature, the Cu substrate shrinks a lot while diamond has much less contractions [21]. In the case of a diamond coating on the bare copper surface, a high stress will occur between these two layers and will result in cracks in the diamond coatings. Part of the diamond coating may peel off from the substrate due to the high stress and low affinity between

diamond and Cu [14]. In Fig. 3(A), some cracks formed on the diamond coatings on the bare Cu substrate can be observed. These cracks also resulted from the stress between diamond coatings and the Cu substrate. On comparing SEM images of the diamond coating on the bare and the sandblasted copper, we can infer that the sandblasting treatment improved the adhesion between the diamond and the Cu substrate. This is due to mechanical interlocking between copper surface and diamond coating [14].

Fig. 4 shows the XRD patterns of the samples presented in Fig. 3. The diamond and Cu have a similar crystal lattice and their main XRD diffraction peaks are very close [22]. The peaks at 50.1° , 59.7° and 89.1° are the diffractions from the (111), (200), and (220) patterns of Cu. The diamond diffraction pattern is weak in the untreated Cu substrate because some of them peel off, and reduce the diffraction intensity. Comparing the untreated to the sandblasted sample, the left shift of the diamond [111] pattern and [220] pattern is because of the stress between the diamond and Cu substrate.

Fig. 5(A and B) shows the SEM images of the CNT coated Cu substrates with and without the diamond intermediate layer. The growth of CNT is controlled by the sputtering time of catalyst preparation and the deposition time of the CNT. The vertically aligned CNTs were achieved by a DC-plasma hot filament CVD process. In the present coated sample, the catalyst's spinning time and deposition time are limited to 90 seconds and 20 minutes respectively. The length and diameter of the CNT coatings can be measured with a highly magnified image as 300–500 nm and 50 nm respectively. Tap water flow test is conducted to find CNT adhesion on copper substrate with a mass flow rate of 10 l/min and the distance between tap and the

Table 1
Surface roughness results.

Surfaces	X direction (μm)	Y direction (μm)	Average (μm)
Untreated Cu	0.22	0.28	0.25
Sand blasted Cu	1.99	2.00	1.95
CNT coated Cu with diamond coating	2.32	2.36	2.34

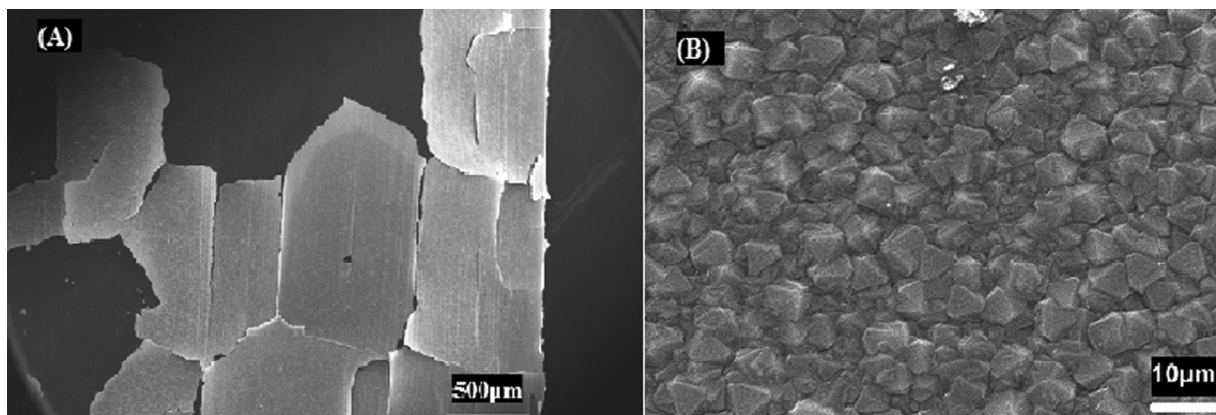


Fig. 3. EM images of diamond coating on the (A) bare copper surface and (B) sand blasted copper.

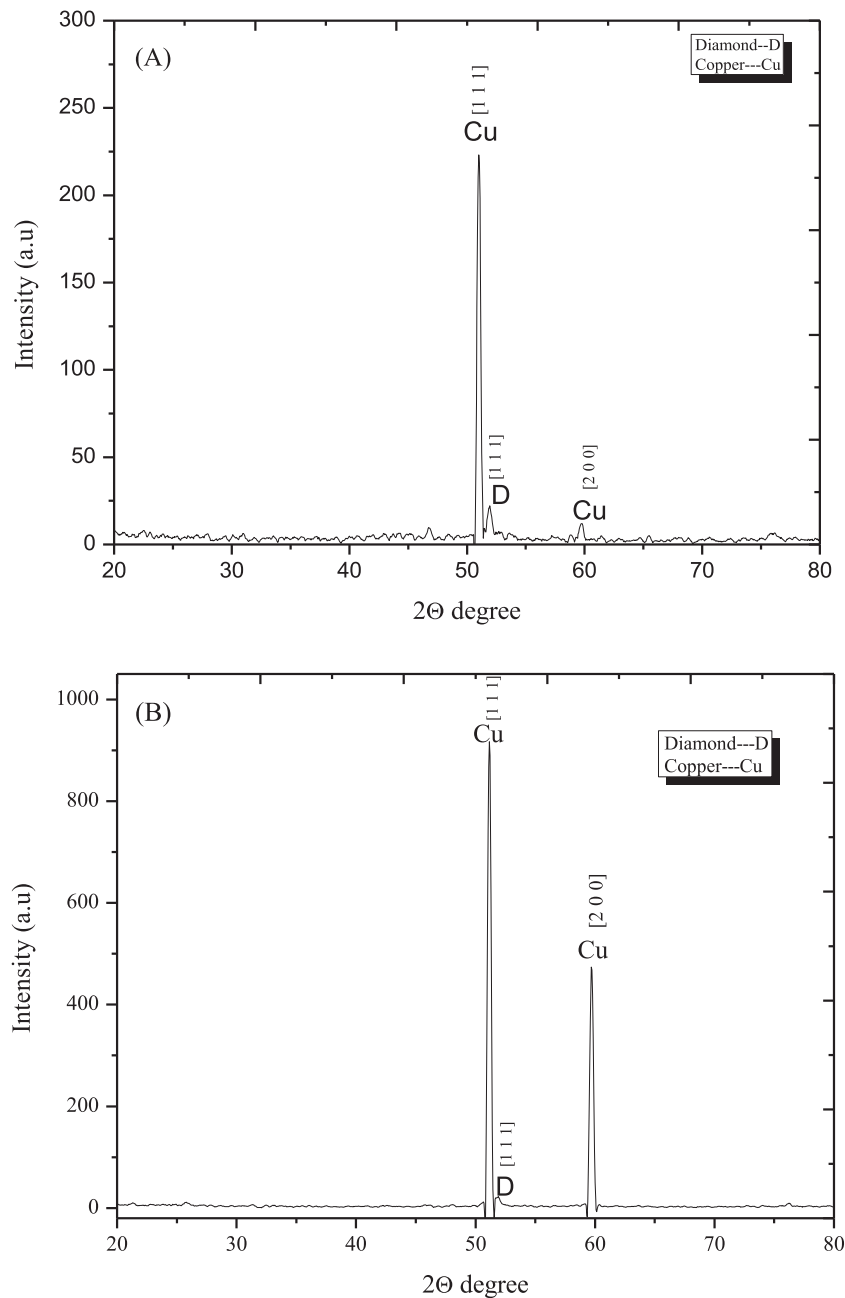


Fig. 4. XRD patterns of the diamond coating on (A) untreated copper and (B) sandblasted copper.

coated surface is maintained around 20 cm. This test is done at room temperature condition. The test is carried out for 20 min. After the tap water flow test, we observed a good adhesion of the CNT on the diamond intermediate layer, compared to that without the intermediate layer. An increase in the adhesion is due to the formation of a carbon bond between the diamond and CNT. This increase in adhesion produces less intermediate resistance between the substrate and coating.

Table 2 shows the variation of surface roughness values of CNT coated surface with and without the diamond intermediate layer for before and after conducting the tap water flow test. The surface roughness of CNT coated Cu without diamond intermediate layer is decreasing as some of the CNTs were washed away from the surface due to poor adhesion.

In Fig. 6(A), a sharp peak centred at 1335 cm^{-1} represents a high quality diamond. The impurity of the diamond is a barrier for thermal

conducting. Thus, this high purity would have a high thermal conductivity. The shift in the diamond peak is due to the stress produced by the mechanical interlock between the diamond and sand blasted copper surface. In Fig. 6(B), the peaks centred at 1350 cm^{-1} and 1580 cm^{-1} are the peaks of CNT respectively. These peaks confirm the existence of the CNT.

Table 2
Surface roughness variation by tap water flow test

CNT coated Cu surface	Before conducting the test (μm)	After conducting the test (μm)
With diamond coating	2.34	2.26
Without diamond coating	2.14	1.23

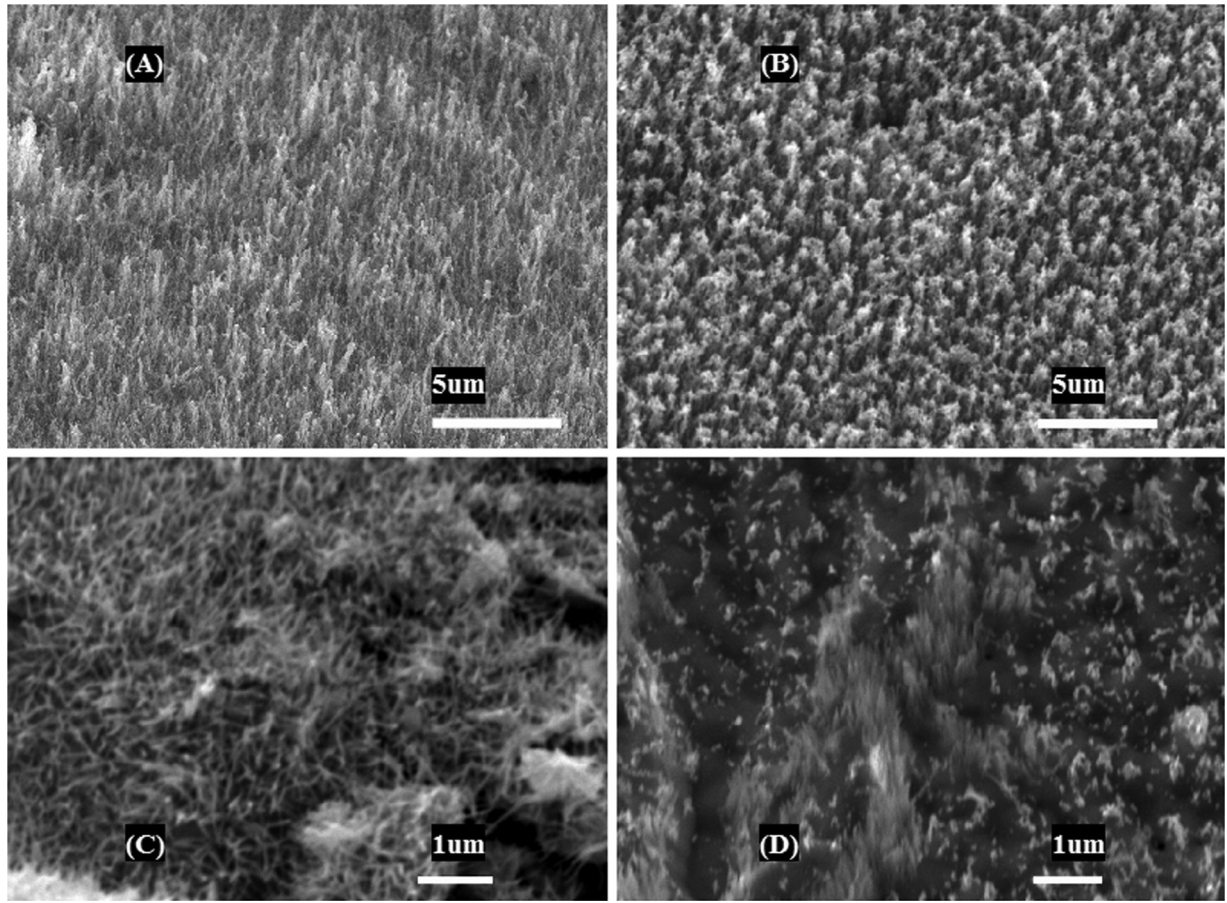


Fig. 5. SEM images of CNT coating (A) on diamond coated Cu, (B) on bare Cu, (C) on diamond coated Cu after tap water test, and (D) on bare copper after tap water test.

3.2. Validation of experimental setup

To validate the experimental setup, experiments were conducted with de-mineralized water as the working fluid. Experiments were conducted with bare copper as the test plates, and the obtained results were compared with the equation proposed by the Rohsenow [23].

$$q = \mu_l h_{fg} \left[\frac{g(\rho_l - \rho_v)}{\sigma} \right]^{\frac{1}{2}} \left[\frac{C_{pl}(T_s - T_{sat})}{C_{sf} h_{fg} Pr_l^n} \right]^{-3} \quad (5)$$

From Fig. 7, we can conclude that the obtained experimental results were comparable with the theoretical values. The maximum error percentage obtained was around 9.5%. As there is not much difference between the theoretical and the experimental values obtained, the test section is considered to be valid.

3.3. Pool boiling heat transfer studies

Pool boiling experiments were conducted with an initial bulk temperature of de-mineralized water at 90 °C. The pool boiling heat flux data of CNT coated Cu and sand blasted Cu were compared with that of the bare Cu. During the experimentation, we found that the commencement of nucleation for the CNT coated surface and sand blasted surface occurred at a lower surface temperature and lower heat flux as compared to that of the bare copper surface. The nucleation sites for the CNT coated surface were more due to its higher relative roughness compared to the sand blasted copper. The higher nucleation density will lead to an early onset of boiling;

thereby, wall superheat of the sand blasted surface decreases for corresponding heat flux considered with the bare copper surface. Larger heat fluxes were obtained from the sand blasted copper and CNT coated surface with respect to wall superheat. The transition from nucleate to film boiling over sand blasted copper and bare copper is rapid compared to the CNT coated surface. In the case of the CNT coated surface, the initial bubble density was higher due to its higher surface roughness, but its hydrophobicity slowed down the growth of the formed bubbles. At higher heat flux, the bubbles spread over the surface, causing coalescence [5], which leads to a slow transition from nucleate to film boiling. Another major reason for the increase in the bubble nucleation over the CNT coated surface is the protruding nanostructures. This in turn increases the effective surface area near the wall, leading to increase in the nucleation sites.

3.4. Heat flux data

The amount of heat flux supplied to the water using the copper heating block was calculated using Eqn. (1). Fig. 8 shows the saturated pool boiling curves for the three different surfaces (bare copper surface, sandblasted copper surface, and CNT-coated surface). The data were obtained from the average of the temperature measurements at the three thermocouple locations and the initial bulk temperature of the fluid.

The shapes of the boiling curves for all the above mentioned surfaces were almost the same but a leftward temperature shift occurred in the case of the CNT coated surface for different heat fluxes. From the boiling curves, we can infer that in the CNT coated surface the

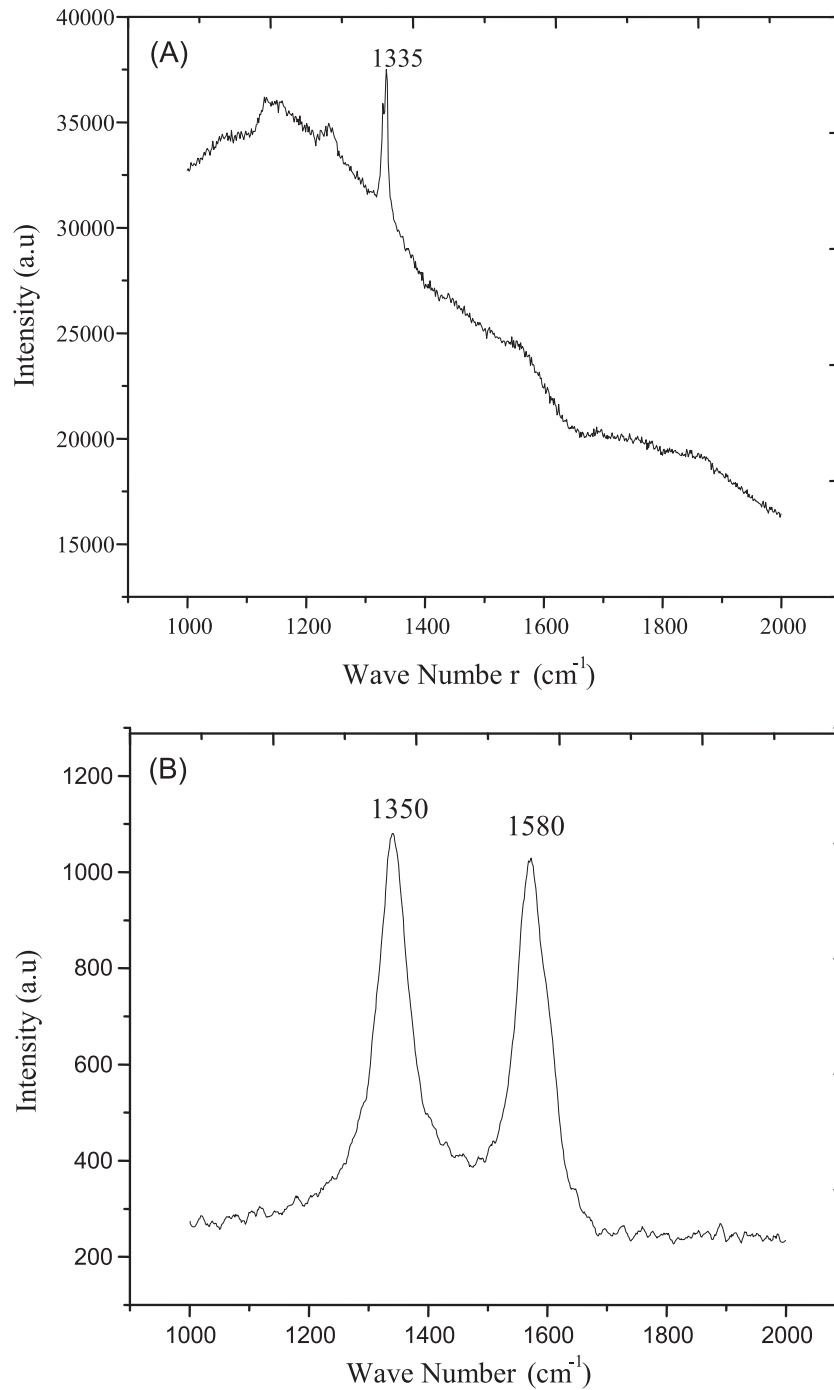


Fig. 6. Raman spectrum of (A) diamond coating on sand blasted Cu. (B) CNT coating on the diamond coated Cu.

wall superheat is less than that of the other two samples for all the measured conditions. This anomalous behaviour is due to hydrophobicity, fin action, increasing surface area and high relative surface roughness. For the sandblasted copper, in a low heat flux region, there is a reduction in the wall superheat and this effect was close enough to CNT coated sample. Close to the critical heat flux region, the wall superheat of the bare copper and sand blasted copper is almost same. In the case of sand blasted copper, due to its higher relative roughness, the initial nucleation starts immediately and a hike in CHF was also found due to the higher heat transfer area. It is observed that the CNT coated surface enhances the heat flux in

the nucleate boiling regime by 48%, and that of sand blasted copper is 12%, with respect to the bare copper surface.

3.5. Critical heat flux

Table 3 shows the variation of the critical heat flux and surface roughness of different tested samples. Both the CNT coated surface and sand blasted surface shows an enhancement in the CHF with respect to the bare copper surface. The enhancement in the CHF compared to bare copper, for the sand blasted copper, is around 8.5%

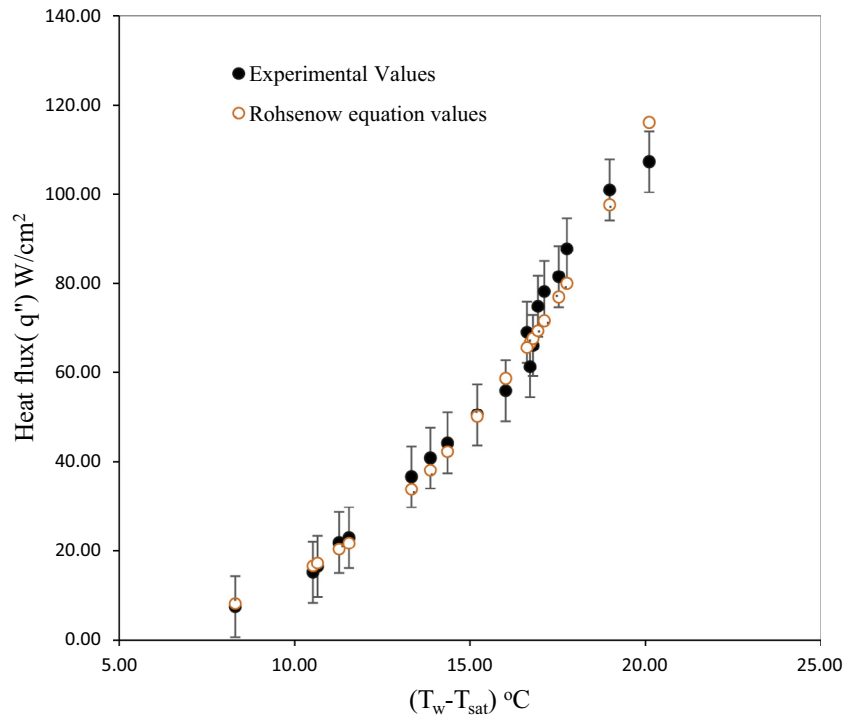


Fig. 7. Comparison of experiment pool boiling results obtained for the bare copper with the results obtained for the proposed equation.

and that of CNT coating is 38%. From Table 1, we can compare the surface roughness of all the three surfaces. The CNT coated surface gives a maximum increase in the surface roughness, compared to other two surfaces.

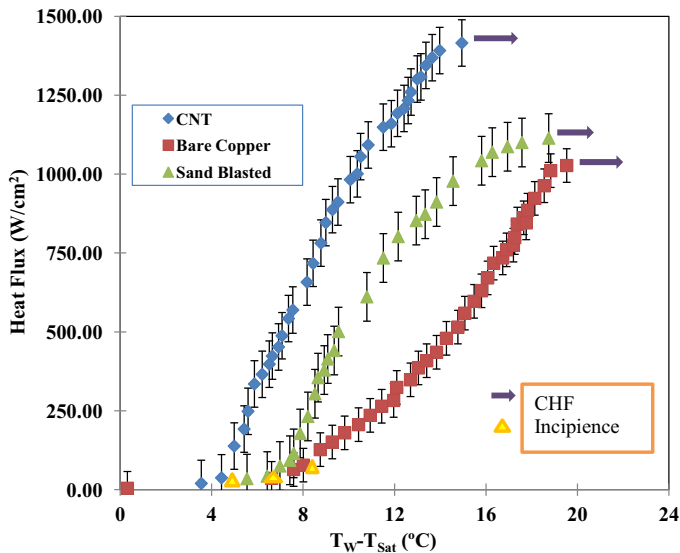


Fig. 8. Pool boiling curve for different surfaces.

Table 3
Measured CHF values of the tested surfaces.

Si. no	Tested surface	CHF (kW/m ²)
1	CNT coated Cu	1.416
2	Sandblasted copper	1.114
3	Bare copper	1.027

CNTs maintained in a vertical direction not only increase the surface area but also increase the fin effectiveness (nano-fin effect) because copper substrate has lower thermal conductivity than the axial thermal conductivity of individual vertically aligned CNTs.

AHN et al. [6] suggested that nano fin effect caused to enhance local quenching of the heater surface due to which an increment in the CHF takes place in the vertically aligned CNTs coated surface. The surface roughness of the sand blasted surface is higher than that of the bare copper surface. An increase in the surface roughness increases the total area for heat transfer which will enhance the CHF of the coated surface as compared to the bare surface.

3.6. Boiling heat transfer coefficient

The effective heat transfer coefficient is calculated using Eqn. (2) up to the critical heat flux. Fig. 9 presents the variation of the boiling heat transfer coefficient with heat flux, for three different samples. The CNT coatings show a very large enhancement in the heat transfer coefficient compared to the other two samples. The maximum increase of 80% in heat transfer coefficient is obtained from the CNT coated surface compared to the bare copper surface. The higher heat transfer coefficient of the CNT coated surface is due to its high surface roughness, effective heat transfer area and Nano fin effect. The use of an intermediate diamond coating reduces the intermediate resistance of the CNT coating over the copper surface by increasing the adhesion. The diamond provides the carbon bonding between the CNT and copper. A reduction in the intermediate resistance leads to a higher heat transfer rate by maintaining a constant temperature between the top surface and the tip of the fin. For the sand blasted copper sample, the boiling heat transfer coefficient is slightly higher than that of the bare Cu due to its higher surface roughness. An increase in the surface roughness increases the effective surface area for heat transfer which in turn increases the heat transfer rate from the surface.

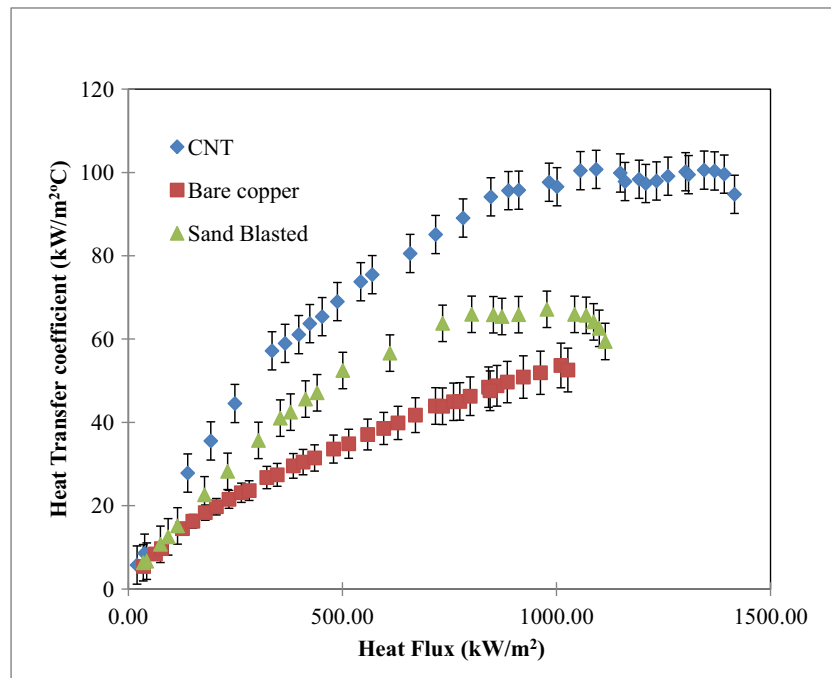


Fig. 9. Variation of the boiling heat transfer coefficient for different surfaces.

3.7. Wettability effects

The wetting ability of water on different surfaces is evaluated by measuring the contact angle of the water on the surfaces. Fig. 10 shows the contact angle on the bare copper, sand blasted Cu, and CNT coatings. It can be seen from Fig. 10 that the contact angles of

the bare copper and sand blasted copper are less than 90°; hence, it is hydrophilic in nature. While the CNT coated surface that provides a contact angle of 133° is hydrophobic. The hydrophobicity of the CNT coated surface is due to the combined effect of diamond intermediate and CNT coating. The wettability is the highest for the bare Cu and the lowest for the CNT coatings. Boiling begins earlier

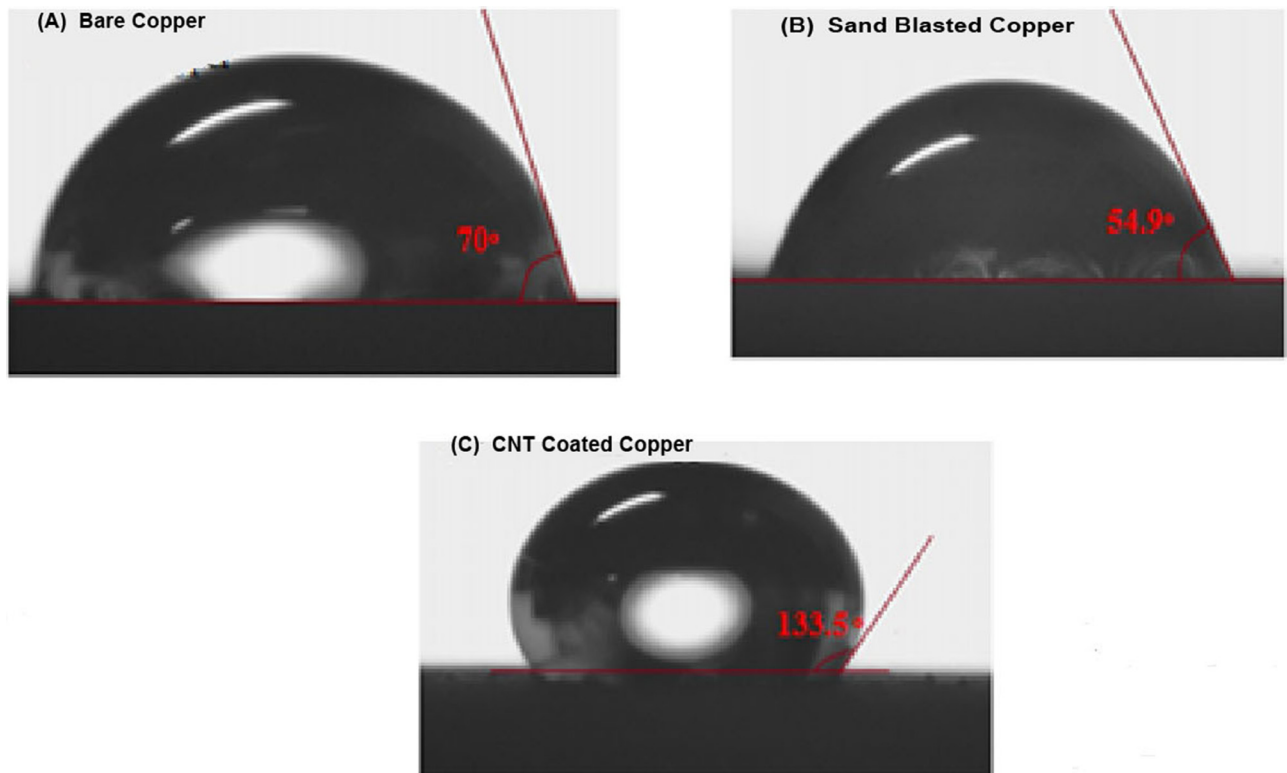


Fig. 10. Static contact angle of (a) bare copper, (b) sand blasted copper, and (c) CNT coated copper substrates.

for the CNT than the other two surfaces due to the early detachment of bubbles from the surfaces. This would enhance the boiling heat transfer performance of the CNT coated surfaces in the low heat flux region.

As the heat flux increases, the bubble density increases, which reduces the boiling heat transfer of copper surface without coating, but this phenomenon was not occurring on the CNT coated surface. The fin action of CNT hinders the growth of the bubbles. This action of the fins can enhance the pool boiling heat transfer characteristics of the CNT coated surface in the high heat flux region.

4. Conclusions

The pool boiling heat flux has been measured and analyzed on the sandblasted Cu, CNT coated copper and bare copper surfaces under an initial bulk temperature of de-mineralized water at 90 °C. The key findings from the study are as follows:

1. Early onset of boiling is obtained for both the CNT and sand blasted Cu samples compared to the bare Cu sample due to higher nucleation density. This effect is due to the hydrophobicity and high relative roughness for the CNT coated samples. Sand blasted copper shows an early onset only due to the high relative roughness.
2. The boiling behaviour is almost similar for all the tested samples. In the case of the CNT, a leftward shift in boiling is obtained compared to the sand blasted copper and bare copper surfaces. For the CNT coated sample, the leftward shift in the boiling curve at low heat flux is due to high nucleation density resulting from its hydrophobic behaviour.

In the case of the CNT coated surface, an increase and leftward shift of the CHF are found due to the fin effect and increase in surface area; thereby, the boiling curve of the CNT coated sample continues along the left side of the sand blasted copper curve and bare copper curve.

For the CNT coated and sand blasted copper surfaces compared with the bare copper, an enhancement in the CHF of 38% and 8.5% respectively were obtained. An enhancement in the heat transfer coefficient of 80% and 13%, respectively, was found in the CNT coated and sand blasted surfaces, compared with the bare copper surface. Increases in the CHF and heat transfer coefficient are mainly due to the increase in the effective surface area for heat transfer. The CNT coated surface shows the better enhancement due to its fin action, high thermal conductivity, and lower intermediate resistance between the CNT and the surface with an intermediate diamond coating.

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Nomenclature

A_c	Area of cross section [m^2]
C_{pl}	Specific heat of the liquid [$J/kg \text{ } ^\circ C$]
C_{sf}	Experimental constant depends on surface fluid combinations

Vap	Vapourization
G	Mass velocity [$kg/m^2 \text{ s}$]
h	Boiling heat transfer coefficient [$W/m^2 \text{ K}$]
h_{fg}	Enthalpy of vaporization [J/kg]
k	Thermal conductivity [$W/m \text{ K}$]
q	Heat flux [W/m^2]
T	Temperature [$^\circ C$]
T_1, T_2	Temperature measured from heater section in vertical direction [$^\circ C$]
T_f	Average fluid temperature [$^\circ C$]
T_{sat}	Saturated temperature of the fluid [$^\circ C$]
T_s	Average surface temperature [$^\circ C$]

Greek symbols

Δx	Vertical distance between thermocouples [m]
μ	Dynamic viscosity of fluid [$N\text{-s}/m^2$]
ρ	Density [kg/m^3]
ω	Uncertainty
σ	Surface tension of the liquid vapour interface [N/m]

Subscripts

Tot	Total
Sub	Subcooled
v	Vapour
w	Bottom wall
act	Actual
c	Cross section
f	Fluid
in	Inlet
l	Liquid
n	Experimental constant depends on the fluid
out	Outlet

Abbreviations

CHF	Critical heat flux [W/cm^2]
CNT	Carbon nanotube
DC	Direct current
DLC	Diamond like carbon
Cu	Copper
HFCD	Hot filament chemical vapour deposition
MEMS	Micro electro mechanic system
ONB	Onset of nucleate boiling
SEM	Scanning electron microscope
XRD	X-ray diffraction
sccm	Standard cubic centimetres per minute
CVT	constant voltage transformer

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